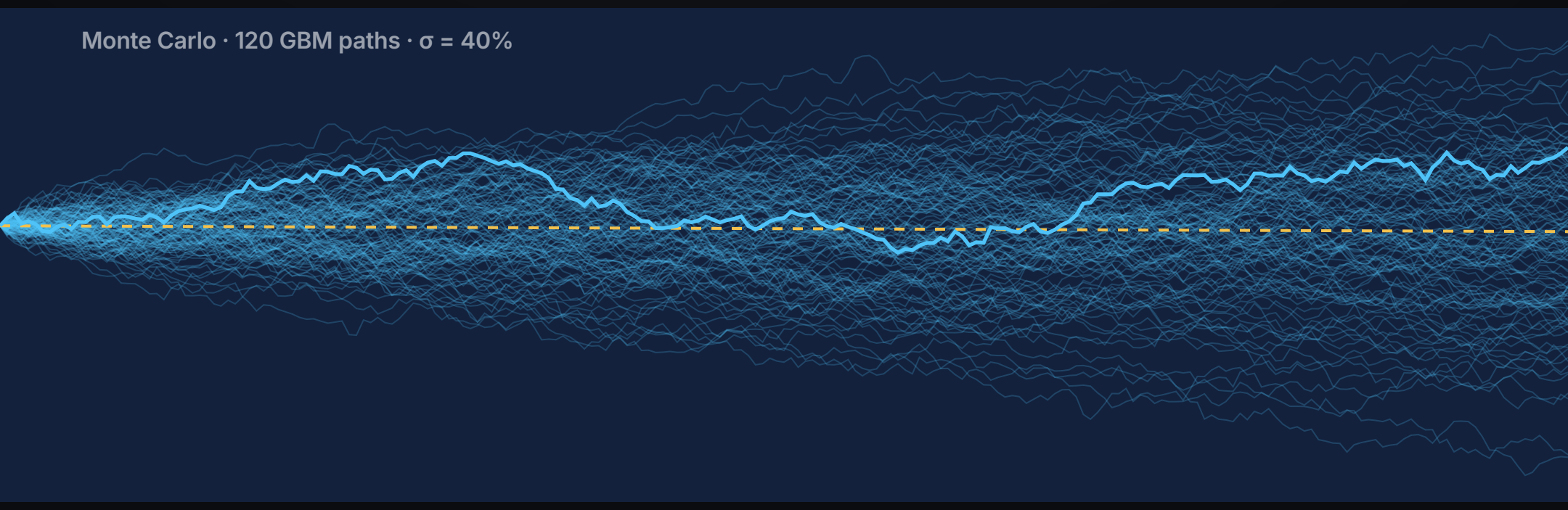
$$\mathrm{VaR}_\alpha = \inf\{\ell : P(L > \ell) \le 1 - \alpha\}$$

$$\sigma_p^2 = \mathbf{w}^\top \Sigma \, \mathbf{w}$$

$$S_T = S_0 \, e^{(\mu - \sigma^2/2)T + \sigma W_T}$$

$$\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha]$$



The 3 ways to compute

VALUE AT RISK

and what it misses

$$\mathrm{VaR}_h = \mathrm{VaR}_1 \sqrt{h}$$

$$P(L > \mathrm{VaR}_\alpha) = 1 - \alpha$$

$\alpha = 99\%$

$h = 1\ \text{day}$

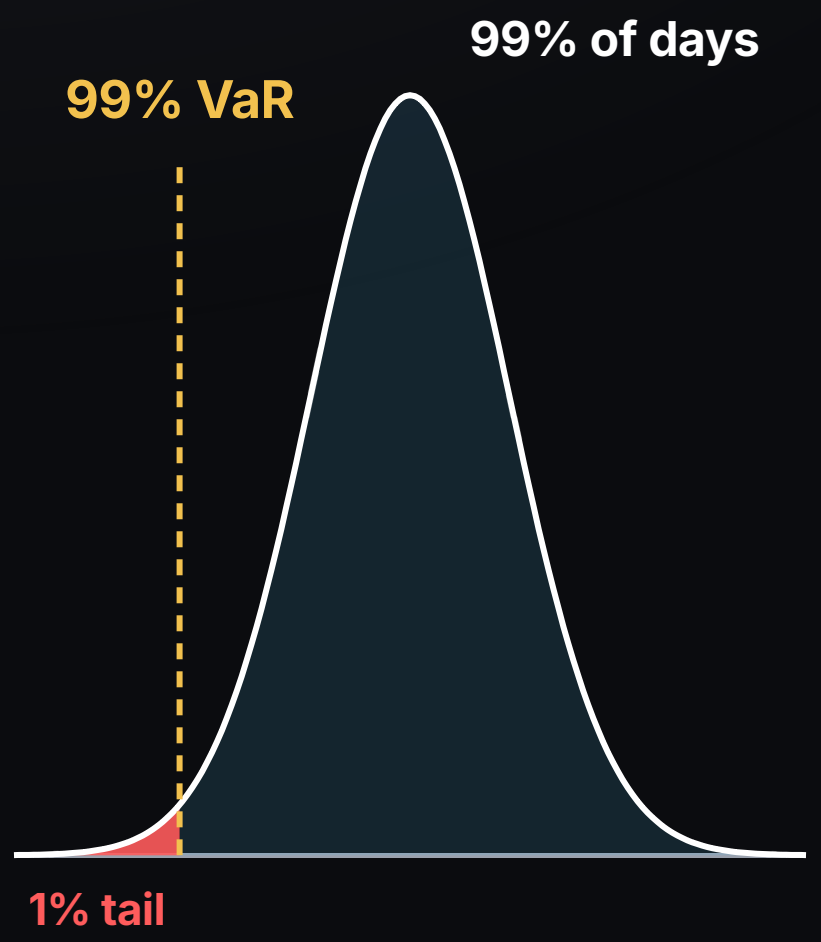
WHAT VaR SAYS

One number, one confidence level, one horizon

VaR answers a single question: over the next day, what loss will we not exceed with 99% confidence? It is a quantile of the P&L distribution, nothing more.

$$P(L > \text{VaR}_\alpha) = 1 - \alpha$$

A 1-day 99% VaR of $279k means losses should exceed $279k on about 1 day in 100. It says nothing about how large those bad days get.



$$L = -\Delta V$$

$$\sigma_p = \sqrt{\mathbf{w}^\top \Sigma \, \mathbf{w}}$$

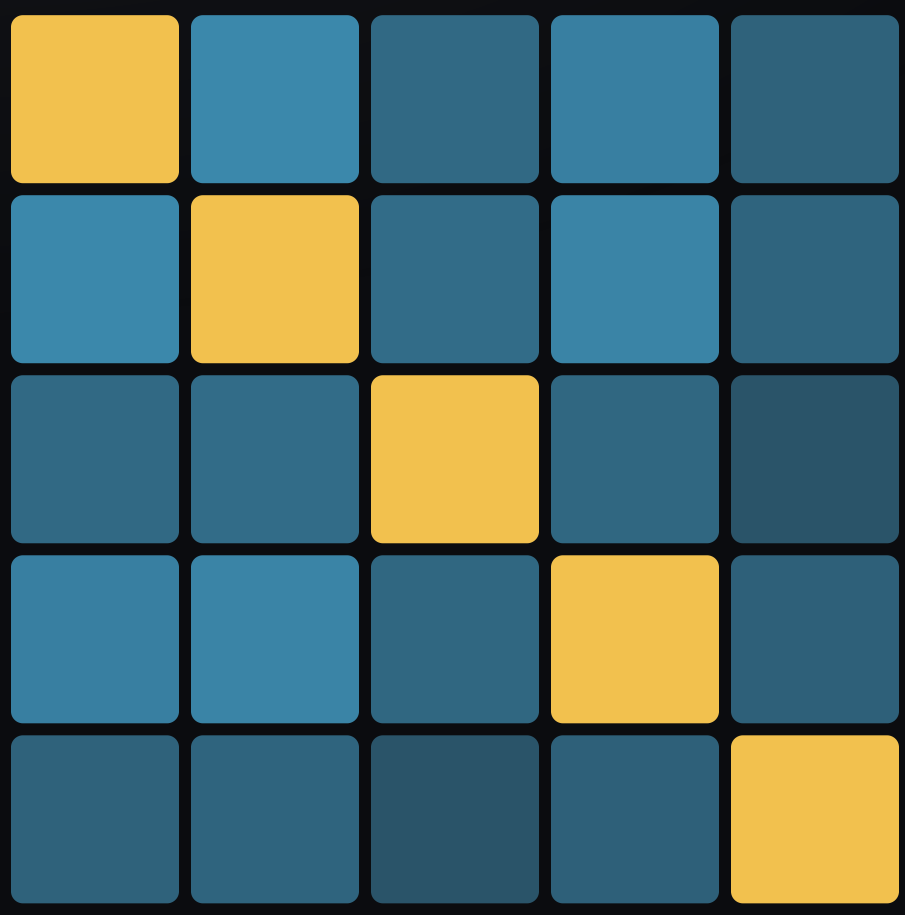
$$z_{0.99} = 2.326$$

1. PARAMETRIC

Variance–covariance method

Assume returns are normal. VaR is then volatility scaled by a z-score: portfolio σ comes from the covariance matrix and your position weights.

$$\mathrm{VaR}_{99\%} = z_{0.99}\, \sigma_p \, V$$



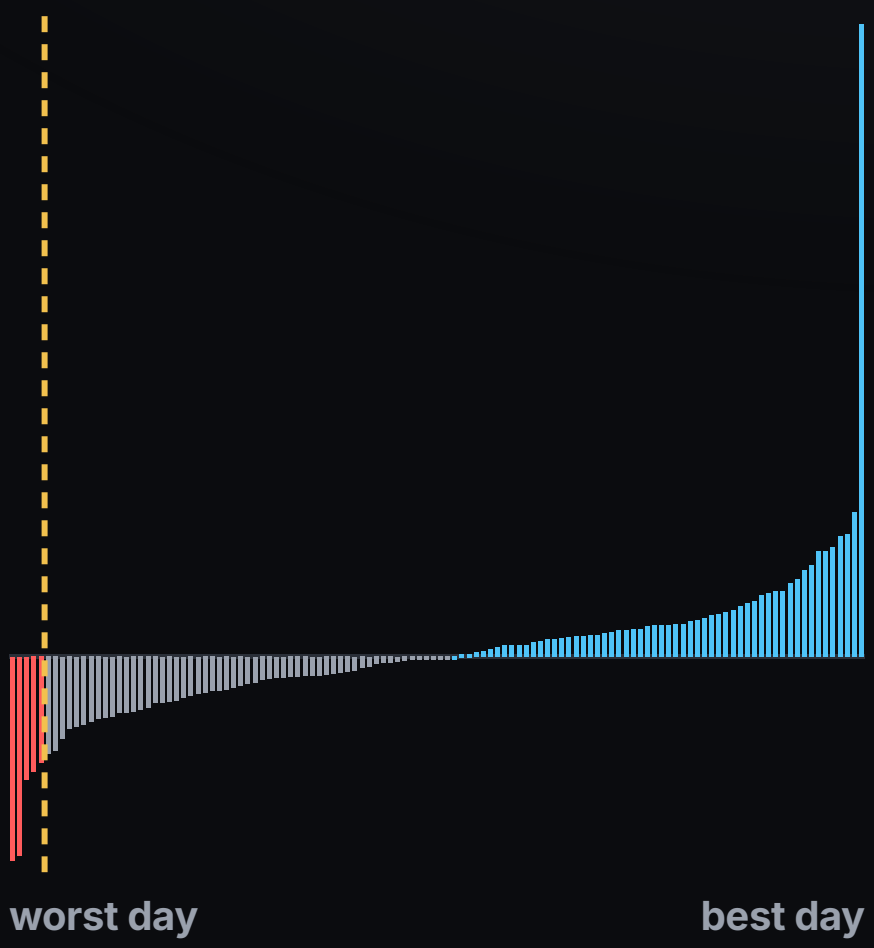
$10M portfolio, 1.2% daily σ: 2.326 × 1.2% × $10M ≈ $279k. Fast and analytic, but it inherits the normal distribution's thin tails.

$$\Sigma = \mathbb{E}[(\mathbf{r} - \boldsymbol{\mu})(\mathbf{r} - \boldsymbol{\mu})^\top]$$

$N = 500$

$500 \times 1\% = 5$

2. HISTORICAL



Historical simulation

No distribution assumed. Replay the last 500 days of market moves on today's portfolio, sort the P&Ls, and read off the loss at the cut-off.

$$\mathrm{VaR}_{99\%} \approx L_{(5)}$$

At 99% with 500 scenarios, VaR is about the 5th-worst loss. Real fat tails are captured, but only the ones in your window: a calm 500 days gives a calm VaR.

$$\hat{F}(x) = \tfrac{1}{N} \sum_i \mathbf{1}\{r_i \le x\}$$

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$$

$$W_t - W_s \sim \mathcal{N}(0,\, t-s)$$

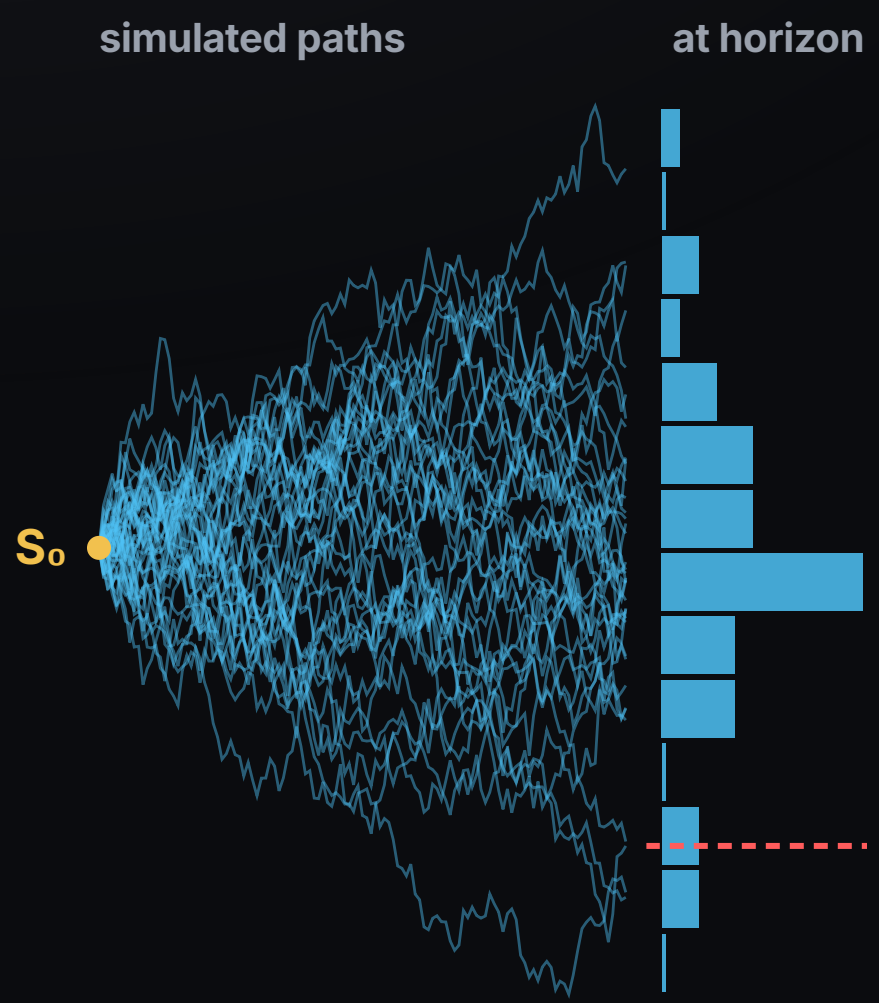
3. MONTE CARLO

Simulate the future instead of replaying the past

Choose a model for every risk factor, simulate thousands of scenarios, reprice the portfolio in each, and take the loss quantile of the simulated P&L.

$$S_T = S_0\, e^{(\mu - \frac{\sigma^2}{2})T + \sigma W_T}$$

It handles options and other non-linear payoffs that the parametric method misses. The price: compute time, and answers only as good as the model.



$$\mathrm{VaR} = -\hat{q}_{1\%}(\Delta V)$$

$$\mathbb{E}[\text{exceptions}] = 250 \times 1\% = 2.5$$

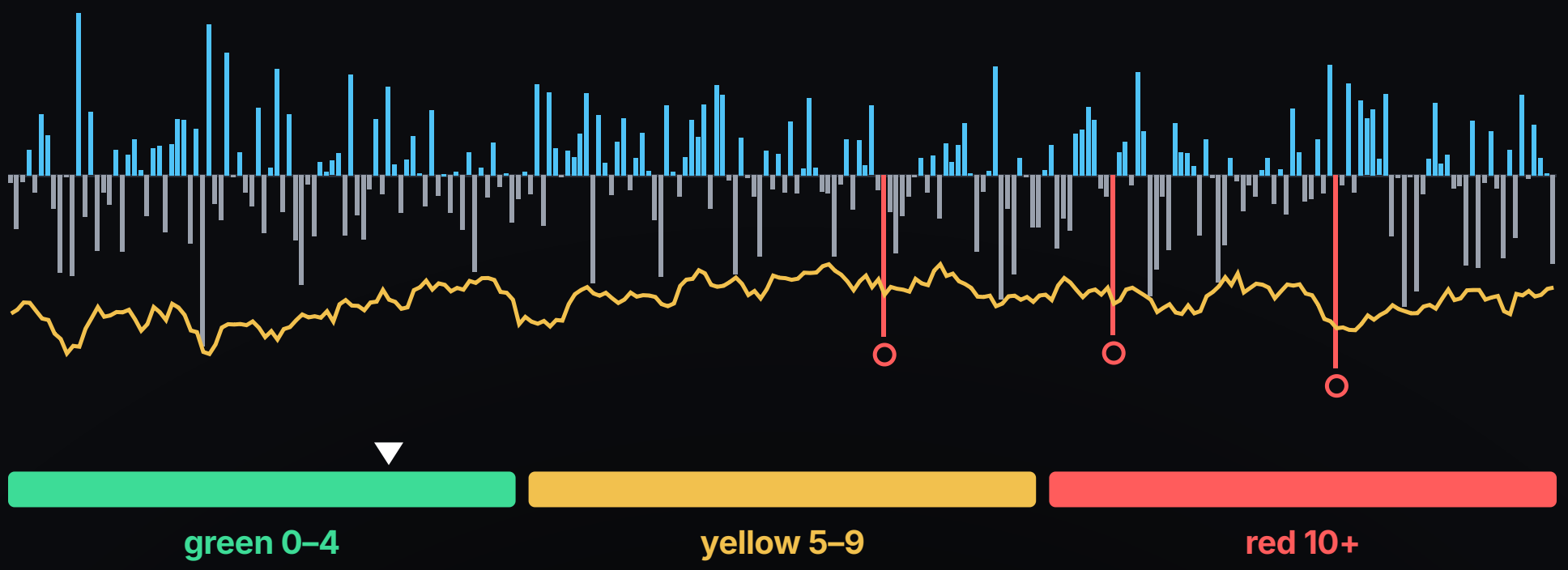
$$\text{VaR}_{10d} = \text{VaR}_{1d}\sqrt{10}$$

SCALE & BACKTEST

√time scaling and the Basel traffic light

Regulators asked for 10-day VaR, usually scaled from 1-day: $279k × √10 ≈ $883k. That holds only if daily returns are independent and identically distributed.

Then test the model: over 250 days a 99% VaR should be breached about 2.5 times. Basel counts the exceptions: 0–4 green, 5–9 yellow, 10 or more red.



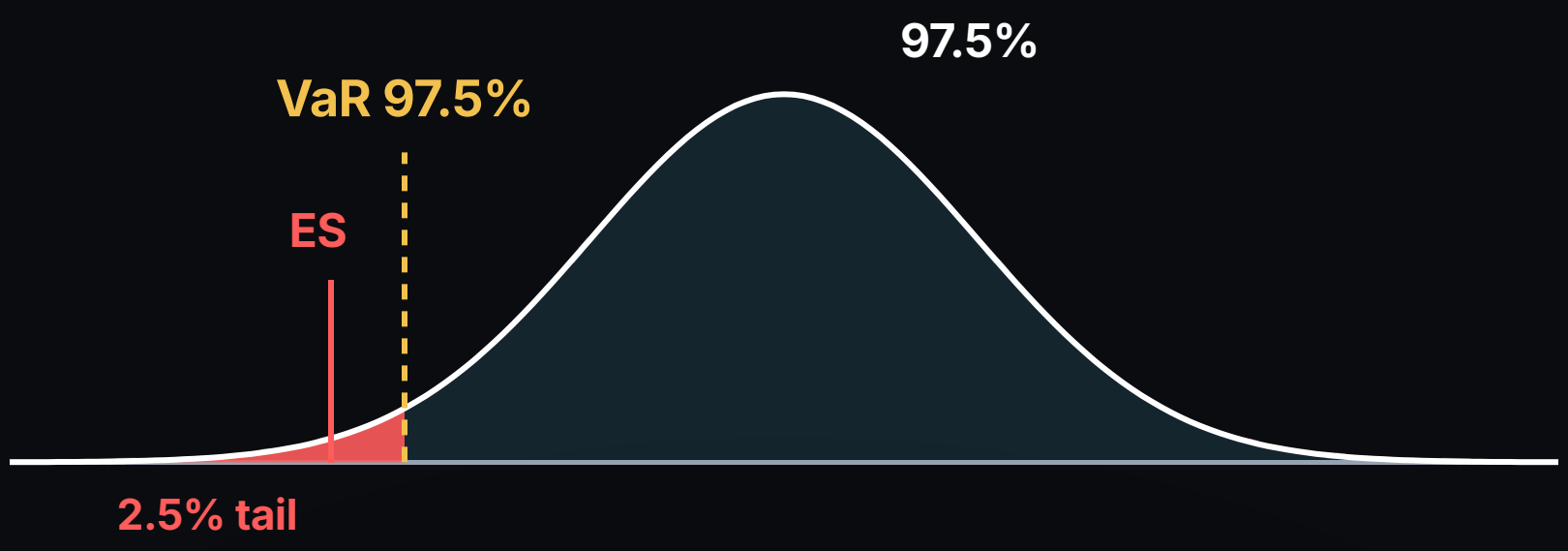
$$\mathrm{ES}_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u\, du$$

$$\mathrm{ES}_{97.5\%} \approx \mathrm{VaR}_{99\%}$$

WHAT VaR MISSES

VaR marks where the tail starts, not how deep it goes. Two portfolios can share a VaR while one hides a far worse tail.

Expected Shortfall averages the losses beyond VaR. Basel's FRTB swapped 99% VaR for 97.5% ES: equal-ish under normality (2.338σ vs 2.326σ), but ES grows when tails get fat.



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Figures assume normally distributed daily P&L; real P&L has fatter tails.